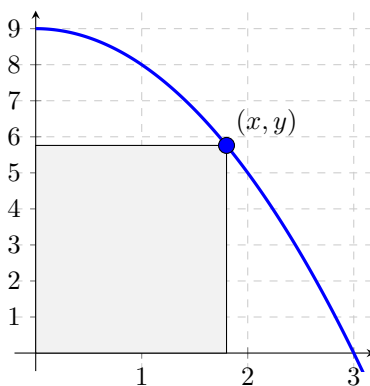


Problem 4

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Problem 1 A rectangle in the first quadrant is constructed by taking a point (x, y) on the graph of the function $f(x) = 9 - x^2$, drawing a line segment vertically downward to the x -axis and a line segment horizontally leftward to the y -axis, as in the picture below. Denote the **length of the base** (along the x -axis) of the rectangle by x , the **height of the rectangle** (along the y -axis) by y (in meters), and the **PERIMETER** of the rectangle by P . When $x = 2$ m (and only at that moment), the height y is shrinking at a rate of $\frac{1}{5}$ m/s. Find the value of $\left[\frac{dP}{dt} \right]_{x=2m}$ **by performing the steps below.**



- (a) **Find a formula** for the height, y , of the rectangle as a function of x . (This is denoted as $y(x)$.)

Explanation. The height of the rectangle is the y -coordinate of the point in the upper-right corner. Since this point lies on the graph of $f(x) = 9 - x^2$, we know $y = 9 - x^2$.

- (b) **Find the value** of $\left[\frac{dx}{dt} \right]_{x=2m}$.

Explanation. Since $y = 9 - x^2$, we know $\frac{dy}{dt} = -2x \frac{dx}{dt}$. Solving for $\frac{dx}{dt}$

gives:

$$\begin{aligned}\frac{dx}{dt} &= -\frac{\frac{dy}{dt}}{2x} \\ \left[\frac{dx}{dt}\right]_{x=2m} &= -\frac{\left[\frac{dy}{dt}\right]_{x=2m}}{2(2)} \\ &= -\frac{(-1/5)}{4} \\ &= \frac{1}{20}\end{aligned}$$

- (c) **Find a formula** for the perimeter, P , of the rectangle as a function of x . (This is denoted as $P(x)$.)

Explanation.

$$\begin{aligned}P(x) &= 2x + 2y \\ &= 2x + 2(9 - x^2) \\ &= -2x^2 + 2x + 18\end{aligned}$$

- (d) **Find the value** of $\left[\frac{dP}{dt}\right]_{x=2m}$.

Explanation.

$$\begin{aligned}P(x) &= -2x^2 + 2x + 18 \\ \frac{dP}{dt} &= (-4x + 2)\frac{dx}{dt} \\ eval \frac{dP}{dt} \bigg|_{x=2m} &= \left[(-4x + 2)\frac{dx}{dt}\right]_{x=2m} \\ &= (-4(2) + 2)\left(\frac{1}{20}\right) \\ &= -\frac{3}{10}\end{aligned}$$

- (e) **Write a sentence to explain** what the value found in (d) means about the rectangle. (Don't forget UNITS.)

Explanation. At the instant when $x = 2$ m, the perimeter of the rectangle is shrinking at the rate of $\frac{3}{10}$ m/s.